

SN74AVC8T245 8-Bit Dual-Supply Bus Transceiver With Configurable Voltage Translation and 3-State Outputs

1 Features

- Latch-up performance exceeds 100 mA per JESD 78, Class II
- ESD protection exceeds JESD 22:
 - 8000-V Human-Body Model (A114-A)
 - 200-V Machine Model (A115-A)
 - 1000-V Charged-Device Model (C101)
- Control inputs V_{IH}/V_{IL} levels are referenced to V_{CCA} voltage
- V_{CC} isolation feature – if either V_{CC} input is at GND, all I/O ports are in the high-impedance state
- I_{off} supports partial power-down mode operation
- Fully configurable dual-rail design allows each port to operate over the full 1.4-V to 3.6-V power-supply range
- I/Os are 4.6-V tolerant
- Maximum data rates:
 - 170Mbps ($V_{CCA} < 1.8\text{ V}$ or $V_{CCB} < 1.8\text{ V}$)
 - 320Mbps ($V_{CCA} \geq 1.8\text{ V}$ and $V_{CCB} \geq 1.8\text{ V}$)

2 Applications

- [Personal electronic](#)
- [Industrial](#)
- [Enterprise](#)
- [Telecom](#)

3 Description

This 8-bit noninverting bus transceiver uses two separate configurable power-supply rails. The SN74AVC8T245 is optimized to operate with V_{CCA}/V_{CCB} set at 1.4 V to 3.6 V. The device is operational with V_{CCA} and V_{CCB} as low as 1.2 V. The A port is designed to track V_{CCA} . V_{CCA} accepts any supply voltage from 1.2 V to 3.6 V. The B port is designed to track V_{CCB} . V_{CCB} accepts any supply voltage from 1.2 V to 3.6 V. This allows for universal low-voltage bidirectional translation between any of the 1.2-V, 1.5-V, 1.8-V, 2.5-V, and 3.3-V voltage nodes.

The SN74AVC8T245 is designed for asynchronous communication between data buses. The device transmits data from the A bus to the B bus or from the B bus to the A bus, depending on the logic level at the direction-control (DIR) input. The output-enable ($\overline{OE}$) input can be used to disable the outputs so the buses are effectively isolated.

The SN74AVC8T245 is designed so that the control pins (DIR and $\overline{OE}$) are supplied by V_{CCA} .

The SN74AVC8T245 is compatible with a single-supply system and can be replaced later with a '245 function, with minimal printed circuit board redesign.

This device is fully specified for partial-power-down applications using I_{off} . The I_{off} circuitry disables the outputs, thus preventing damaging current backflow through the device when it is powered down.

The V_{CC} isolation feature allows both ports to be in the high-impedance state when either V_{CC} input is at GND.

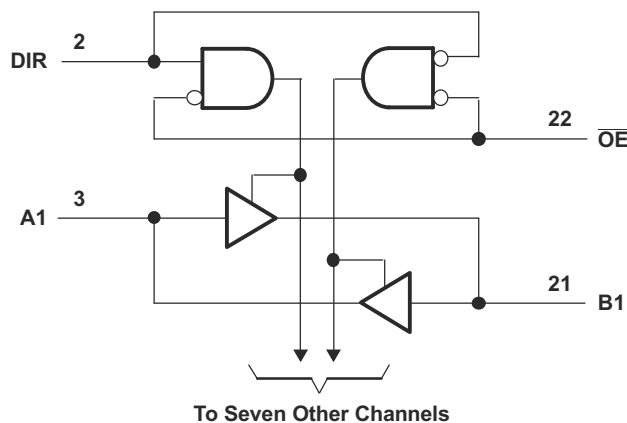
To put the device into the high-impedance state during power up or power down, tie $\overline{OE}$ to V_{CC} through a pullup resistor; the current-sinking capability of the driver determines the minimum value of the resistor.

Package Information

PART NUMBER	PACKAGE ⁽¹⁾	PACKAGE SIZE ⁽²⁾
SN74AVC8T245	RHL (VQFN, 24)	5.5 mm × 3.5 mm
	PW (TSSOP, 24)	7.8 mm × 6.4 mm
	DGV (TVSOP, 24)	5 mm × 6.4 mm

(1) For more information, see [Section 10](#).

(2) The package size (length × width) is a nominal value and includes pins, where applicable.



Logic Diagram (Positive Logic)



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4 Pin Configuration and Functions

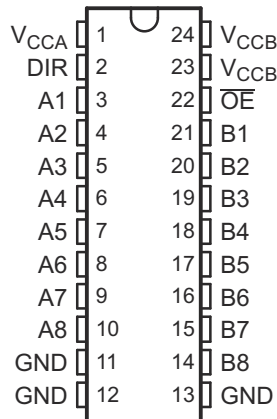


Figure 4-1. DGV or PW Package, 24-Pin TVSOP or TSSOP (Top View)

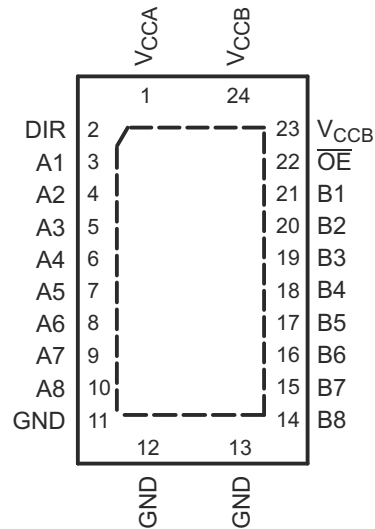


Figure 4-2. RHL Package, 24-Pin VQFN (Top View)

Table 4-1. Pin Functions

PIN		I/O	DESCRIPTION
NAME	NO.		
A1	3	I/O	Input/output A1. Referenced to V_{CCA} .
A2	4	I/O	Input/output A2. Referenced to V_{CCA} .
A3	5	I/O	Input/output A3. Referenced to V_{CCA} .
A4	6	I/O	Input/output A4. Referenced to V_{CCA} .
A5	7	I/O	Input/output A5. Referenced to V_{CCA} .
A6	8	I/O	Input/output A6. Referenced to V_{CCA} .
A7	9	I/O	Input/output A7. Referenced to V_{CCA} .
A8	10	I/O	Input/output A8. Referenced to V_{CCA} .
B1	21	I/O	Input/output B1. Referenced to V_{CCB} .
B2	20	I/O	Input/output B2. Referenced to V_{CCB} .
B3	19	I/O	Input/output B3. Referenced to V_{CCB} .
B4	18	I/O	Input/output B4. Referenced to V_{CCB} .
B5	17	I/O	Input/output B5. Referenced to V_{CCB} .
B6	16	I/O	Input/output B6. Referenced to V_{CCB} .
B7	15	I/O	Input/output B7. Referenced to V_{CCB} .
B8	14	I/O	Input/output B8. Referenced to V_{CCB} .
DIR	2	I	Direction-control signal
GND	11, 12, 13	—	Ground
$\overline{OE}$	22	I	3-state output-mode enables. Pull $\overline{OE}$ high to place all outputs in 3-state mode. Referenced to V_{CCA} .
V_{CCA}	1	—	A-port supply voltage. $1.2\text{ V} \leq V_{CCA} \leq 3.6\text{ V}$
V_{CCB}	23, 24	—	B-port supply voltage. $1.2\text{ V} \leq V_{CCB} \leq 3.6\text{ V}$

5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted) ⁽¹⁾

			MIN	MAX	UNIT
V_{CCA}, V_{CCB}	Supply voltage		−0.5	4.6	V
V_I	Input voltage ⁽²⁾	I/O ports (A port)	−0.5	4.6	V
		I/O ports (B port)	−0.5	4.6	
		Control inputs	−0.5	4.6	
V_O	Voltage range applied to any output in the high-impedance or power-off state ⁽²⁾	A port	−0.5	4.6	V
		B port	−0.5	4.6	
V_O	Voltage range applied to any output in the high or low state ⁽²⁾ ⁽³⁾	A port	−0.5	$V_{CCA} + 0.5$	V
		B port	−0.5	$V_{CCB} + 0.5$	
I_{IK}	Input clamp current	$V_I < 0$		−50	mA
I_{OK}	Output clamp current	$V_O < 0$		−50	mA
I_O	Continuous output current		−50	50	mA
	Continuous current through V_{CCA} , V_{CCB} , or GND		−100	100	mA
T_{stg}	Storage temperature		−65	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) The input voltage and output negative-voltage ratings may be exceeded if the input and output current ratings are observed.

(3) The output positive-voltage rating may be exceeded up to 4.6 V maximum if the output current rating is observed.

5.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±8000	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±1000	
		Machine model (MM)	±200	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

See (1) (2) (3)

			V _{CCI}	V _{CCO}	MIN	MAX	UNIT
V _{CCA}	Supply voltage				1.2	3.6	V
V _{CCB}	Supply voltage				1.2	3.6	V
V _{IH}	High-level input voltage	Data inputs	1.2 V to 1.95 V		V _{CCI} × 0.65		V
			1.95 V to 2.7 V		1.6		
			2.7 V to 3.6 V		2		
V _{IL}	Low-level input voltage	Data inputs	1.2 V to 1.95 V		V _{CCI} × 0.35		V
			1.95 V to 2.7 V		0.7		
			2.7 V to 3.6 V		0.8		
V _{IH}	High-level input voltage	DIR (referenced to V _{CCA})	1.2 V to 1.95 V		V _{CCA} × 0.65		V
			1.95 V to 2.7 V		1.6		
			2.7 V to 3.6 V		2		
V _{IL}	Low-level input voltage	DIR (referenced to V _{CCA})	1.2 V to 1.95 V		V _{CCA} × 0.35		V
			1.95 V to 2.7 V		0.7		
			2.7 V to 3.6 V		0.8		
V _I	Input voltage				0	3.6	V
V _O	Output voltage	Active state			0	V _{CCO}	V
		3-state			0	3.6	
I _{OH}	High-level output current			1.2 V		–3	mA
				1.4 V to 1.6 V		–6	
				1.65 V to 1.95 V		–8	
				2.3 V to 2.7 V		–9	
				3 V to 3.6 V		–12	
I _{OL}	Low-level output current			1.2 V		3	mA
				1.4 V to 1.6 V		6	
				1.65 V to 1.95 V		8	
				2.3 V to 2.7 V		9	
				3 V to 3.6 V		12	
Δt/Δv	Input transition rise or fall rate					5	ns/V
T _A	Operating free-air temperature				–40	125	°C

(1) V_{CCI} is the V_{CC} associated with the input port.

(2) V_{CCO} is the V_{CC} associated with the output port.

(3) All unused data inputs of the device must be held at V_{CCI} or GND to ensure proper device operation. See [Implications of Slow or Floating CMOS Inputs](#), SCBA004.

5.4 Thermal Information

THERMAL METRIC ⁽¹⁾		SN74AVC8T245			UNIT
		DGV	PW	RHL	
		24 PINS	24 PINS	24 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	116.7	93.1	36.8	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	48.5	36.7	32.5	°C/W
R _{θJB}	Junction-to-board thermal resistance	62.1	48.4	15.7	°C/W
ψ _{JT}	Junction-to-top characterization parameter	7.0	93.1	0.7	°C/W
ψ _{JB}	Junction-to-board characterization parameter	61.6	48.0	15.6	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	N/A	N/A	5.6	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

5.5 Electrical Characteristics

over recommended operating free-air temperature range (unless otherwise noted)^{(2) (1)}

PARAMETER	TEST CONDITIONS	V _{CCA}	V _{CCB}	T _A = 25°C			–40°C to +85°C		–40°C to +125°C		UNIT
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
V _{OH}	I _{OH} = –100 μA	V _I = V _{IH}	1.2 V to 3.6 V	1.2 V to 3.6 V			V _{CCO} – 0.2		V _{CCO} – 0.2		V
	I _{OH} = –3 mA		1.2 V	1.2 V	0.95						
	I _{OH} = –6 mA		1.4 V	1.4 V			1.05		1		
	I _{OH} = –8 mA		1.65 V	1.65 V			1.2		1.2		
	I _{OH} = –9 mA		2.3 V	2.3 V			1.75		1.75		
	I _{OH} = –12 mA		3 V	3 V			2.3		2.3		
V _{OL}	I _{OL} = 100 μA	V _I = V _{IL}	1.2 V to 3.6 V	1.2 V to 3.6 V			0.2		0.2		V
	I _{OL} = 3 mA		1.2 V	1.2 V	0.15						
	I _{OL} = 6 mA		1.4 V	1.4 V			0.35		0.35		
	I _{OL} = 8 mA		1.65 V	1.65 V			0.45		0.45		
	I _{OL} = 9 mA		2.3 V	2.3 V			0.55		0.55		
	I _{OL} = 12 mA		3 V	3 V			0.7		0.7		
I _I Control inputs	V _I = V _{CCA} or GND		1.2 V to 3.6 V	1.2 V to 3.6 V	–0.25	±0.025	0.25	–1	1	±1	μA
I _{off} A or B port	V _I or V _O = 0 to 3.6 V		0 V	0 V to 3.6 V	–1	±0.1	1	–5	5	±5	μA
			0 V to 3.6 V	0 V	–1	±0.1	1	–5	5	±5	
I _{OZ} ⁽³⁾ A or B port	V _O = V _{CCO} or GND, V _I = V _{CCi} or GND, OE = V _{IH}		3.6 V	3.6 V		±0.5	±2.5		±5	±5	μA
I _{CCA}	V _I = V _{CCi} or GND, I _O = 0		1.2 V to 3.6 V	1.2 V to 3.6 V				15		15	μA
			0 V	3.6 V				–2		–2	
			3.6 V	0 V				15		15	
I _{CCB}	V _I = V _{CCi} or GND, I _O = 0		1.2 V to 3.6 V	1.2 V to 3.6 V				15		15	μA
			0 V	3.6 V				15		15	
			3.6 V	0 V				–2		–2	
I _{CCA} + I _{CCB}	V _I = V _{CCi} or GND, I _O = 0		1.2 V to 3.6 V	1.2 V to 3.6 V				25		25	μA
C _i Control inputs	V _I = 3.3 V or GND		3.3 V	3.3 V		3.5		4.5			pF
C _{io} A or B port	V _O = 3.3 V or GND		3.3 V	3.3 V		6		7			pF

(1) V_{CCi} is the V_{CC} associated with the input port.

(2) V_{CCO} is the V_{CC} associated with the output port.

(3) For I/O ports, the parameter I_{OZ} includes the input leakage current.

5.6 Switching Characteristics, $V_{CCA} = 1.2\text{ V}$

over recommended operating free-air temperature range, $V_{CCA} = 1.2\text{ V}$ (see [Figure 6-1](#))

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CCB}	$T_A = -40^\circ\text{C to } +85^\circ\text{C}$	$T_A = -40^\circ\text{C to } +125^\circ\text{C}$	UNIT
				TYP	TYP	
t_{PLH}, t_{PHL}	A	B	$V_{CCB} = 1.2\text{ V}$	3.1	3.1	ns
			$V_{CCB} = 1.5\text{ V}$	2.6	2.6	
			$V_{CCB} = 1.8\text{ V}$	2.5	2.5	
			$V_{CCB} = 2.5\text{ V}$	3	3	
			$V_{CCB} = 3.3\text{ V}$	3.5	3.5	
t_{PLH}, t_{PHL}	B	A	$V_{CCB} = 1.2\text{ V}$	3.1	3.1	ns
			$V_{CCB} = 1.5\text{ V}$	2.7	2.7	
			$V_{CCB} = 1.8\text{ V}$	2.5	2.5	
			$V_{CCB} = 2.5\text{ V}$	2.4	2.4	
			$V_{CCB} = 3.3\text{ V}$	2.3	2.3	
t_{PZH}, t_{PZL}	$\overline{OE}$	A	$V_{CCB} = 1.2\text{ V}$	5.3	5.3	ns
			$V_{CCB} = 1.5\text{ V}$			
			$V_{CCB} = 1.8\text{ V}$			
			$V_{CCB} = 2.5\text{ V}$			
			$V_{CCB} = 3.3\text{ V}$			
t_{PZH}, t_{PZL}	$\overline{OE}$	B	$V_{CCB} = 1.2\text{ V}$	5.1	5.1	ns
			$V_{CCB} = 1.5\text{ V}$	4	4	
			$V_{CCB} = 1.8\text{ V}$	3.5	3.5	
			$V_{CCB} = 2.5\text{ V}$	3.2	3.2	
			$V_{CCB} = 3.3\text{ V}$	3.1	3.1	
t_{PHZ}, t_{PLZ}	$\overline{OE}$	A	$V_{CCB} = 1.2\text{ V}$	4.8	4.8	ns
			$V_{CCB} = 1.5\text{ V}$			
			$V_{CCB} = 1.8\text{ V}$			
			$V_{CCB} = 2.5\text{ V}$			
			$V_{CCB} = 3.3\text{ V}$			
t_{PHZ}, t_{PLZ}	$\overline{OE}$	B	$V_{CCB} = 1.2\text{ V}$	4.7	4.7	ns
			$V_{CCB} = 1.5\text{ V}$	4	4	
			$V_{CCB} = 1.8\text{ V}$	4.1	4.1	
			$V_{CCB} = 2.5\text{ V}$	4.3	4.3	
			$V_{CCB} = 3.3\text{ V}$	5.1	5.1	

5.7 Switching Characteristics, $V_{CCA} = 1.5\text{ V} \pm 0.1\text{ V}$

over recommended operating free-air temperature range, $V_{CCA} = 1.5\text{ V} \pm 0.1\text{ V}$ (see [Figure 6-1](#))

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CCB}	$T_A = -40^\circ\text{C to } +85^\circ\text{C}$			$T_A = -40^\circ\text{C to } +125^\circ\text{C}$			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
t_{PLH}, t_{PHL}	A	B	$V_{CCB} = 1.2\text{ V}$	2.7			3.1			ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		5.4	0.5		14.7	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		4.6	0.5		13.3	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		4.9	0.5		13.9	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		6.8	0.5		17.2	

5.7 Switching Characteristics, $V_{CCA} = 1.5 \text{ V} \pm 0.1 \text{ V}$ (continued)

over recommended operating free-air temperature range, $V_{CCA} = 1.5 \text{ V} \pm 0.1 \text{ V}$ (see [Figure 6-1](#))

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CCB}	$T_A = -40^\circ\text{C to } +85^\circ\text{C}$			$T_A = -40^\circ\text{C to } +125^\circ\text{C}$			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
t_{PLH}, t_{PHL}	B	A	$V_{CCB} = 1.2 \text{ V}$		2.6			3.1		ns
			$V_{CCB} = 1.5 \text{ V} \pm 0.1 \text{ V}$	0.5		5.4	0.5		14.7	
			$V_{CCB} = 1.8 \text{ V} \pm 0.15 \text{ V}$	0.5		5.1	0.5		14.2	
			$V_{CCB} = 2.5 \text{ V} \pm 0.2 \text{ V}$	0.5		4.7	0.5		13.5	
			$V_{CCB} = 3.3 \text{ V} \pm 0.3 \text{ V}$	0.5		4.5	0.5		13.2	
t_{PZH}, t_{PZL}	$\overline{OE}$	A	$V_{CCB} = 1.2 \text{ V}$		3.7			5.3		ns
			$V_{CCB} = 1.5 \text{ V} \pm 0.1 \text{ V}$	1.1		8.7	0.5		20.5	
			$V_{CCB} = 1.8 \text{ V} \pm 0.15 \text{ V}$	1.1		8.7	0.5		20.5	
			$V_{CCB} = 2.5 \text{ V} \pm 0.2 \text{ V}$	1.1		8.7	0.5		20.5	
			$V_{CCB} = 3.3 \text{ V} \pm 0.3 \text{ V}$	1.1		8.7	0.5		20.5	
t_{PZH}, t_{PZL}	$\overline{OE}$	B	$V_{CCB} = 1.2 \text{ V}$		4.8			5.1		ns
			$V_{CCB} = 1.5 \text{ V} \pm 0.1 \text{ V}$	1.1		7.6	0.5		18.6	
			$V_{CCB} = 1.8 \text{ V} \pm 0.15 \text{ V}$	1.1		7.1	0.5		17.7	
			$V_{CCB} = 2.5 \text{ V} \pm 0.2 \text{ V}$	1		5.6	0.5		15.1	
			$V_{CCB} = 3.3 \text{ V} \pm 0.3 \text{ V}$	1		5.2	0.5		14.4	
t_{PHZ}, t_{PLZ}	$\overline{OE}$	A	$V_{CCB} = 1.2 \text{ V}$		3.1			4.8		ns
			$V_{CCB} = 1.5 \text{ V} \pm 0.1 \text{ V}$	0.5		8.6	0.5		20.3	
			$V_{CCB} = 1.8 \text{ V} \pm 0.15 \text{ V}$	0.5		8.6	0.5		20.3	
			$V_{CCB} = 2.5 \text{ V} \pm 0.2 \text{ V}$	0.5		8.6	0.5		20.3	
			$V_{CCB} = 3.3 \text{ V} \pm 0.3 \text{ V}$	0.5		8.6	0.5		20.3	
t_{PHZ}, t_{PLZ}	$\overline{OE}$	B	$V_{CCB} = 1.2 \text{ V}$		4.1			4.7		ns
			$V_{CCB} = 1.5 \text{ V} \pm 0.1 \text{ V}$	0.5		8.4	0.5		20	
			$V_{CCB} = 1.8 \text{ V} \pm 0.15 \text{ V}$	0.5		7.6	0.5		18.6	
			$V_{CCB} = 2.5 \text{ V} \pm 0.2 \text{ V}$	0.5		7.2	0.5		17.9	
			$V_{CCB} = 3.3 \text{ V} \pm 0.3 \text{ V}$	0.5		7.8	0.5		18.9	

5.8 Switching Characteristics, $V_{CCA} = 1.8 \text{ V} \pm 0.15 \text{ V}$

over recommended operating free-air temperature range, $V_{CCA} = 1.8 \text{ V} \pm 0.15 \text{ V}$ (see [Figure 6-1](#))

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CCB}	$T_A = -40^\circ\text{C to } +85^\circ\text{C}$			$T_A = -40^\circ\text{C to } +125^\circ\text{C}$			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
t_{PLH}, t_{PHL}	A	B	$V_{CCB} = 1.2 \text{ V}$		2.5			2.5		ns
			$V_{CCB} = 1.5 \text{ V} \pm 0.1 \text{ V}$	0.5		5.1	0.5		14.2	
			$V_{CCB} = 1.8 \text{ V} \pm 0.15 \text{ V}$	0.5		4.4	0.5		13	
			$V_{CCB} = 2.5 \text{ V} \pm 0.2 \text{ V}$	0.5		4	0.5		12.3	
			$V_{CCB} = 3.3 \text{ V} \pm 0.3 \text{ V}$	0.5		3.9	0.5		12.1	
t_{PLH}, t_{PHL}	B	A	$V_{CCB} = 1.2 \text{ V}$		2.5			2.5		ns
			$V_{CCB} = 1.5 \text{ V} \pm 0.1 \text{ V}$	0.5		4.6	0.5		13.3	
			$V_{CCB} = 1.8 \text{ V} \pm 0.15 \text{ V}$	0.5		4.4	0.5		13	
			$V_{CCB} = 2.5 \text{ V} \pm 0.2 \text{ V}$	0.5		3.9	0.5		12.1	
			$V_{CCB} = 3.3 \text{ V} \pm 0.3 \text{ V}$	0.5		3.7	0.5		11.8	

5.8 Switching Characteristics, $V_{CCA} = 1.8\text{ V} \pm 0.15\text{ V}$ (continued)

over recommended operating free-air temperature range, $V_{CCA} = 1.8\text{ V} \pm 0.15\text{ V}$ (see [Figure 6-1](#))

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CCB}	$T_A = -40^\circ\text{C to } +85^\circ\text{C}$			$T_A = -40^\circ\text{C to } +125^\circ\text{C}$			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
t_{PZH}, t_{PZL}	$\overline{OE}$	A	$V_{CCB} = 1.2\text{ V}$	3			3			ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	1		6.8	0.5		17.2	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	1		6.8	0.5		17.2	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	1		6.8	0.5		17.2	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	1		6.8	0.5		17.2	
t_{PZH}, t_{PZL}	$\overline{OE}$	B	$V_{CCB} = 1.2\text{ V}$	4.6			4.6			ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	1.1		8.2	0.5		19.6	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	1		6.7	0.5		17	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		5.1	0.5		14.2	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		4.5	0.5		13.2	
t_{PHZ}, t_{PLZ}	$\overline{OE}$	A	$V_{CCB} = 1.2\text{ V}$	2.8			2.8			ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		7.1	0.5		17.7	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		7.1	0.5		17.7	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		7.1	0.5		17.7	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		7.1	0.5		17.7	
t_{PHZ}, t_{PLZ}	$\overline{OE}$	B	$V_{CCB} = 1.2\text{ V}$	3.9			3.9			ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		7.8	0.5		18.9	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		6.9	0.5		17.3	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		6	0.5		15.8	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		5.8	0.5		15.4	

5.9 Switching Characteristics, $V_{CCA} = 2.5\text{ V} \pm 0.2\text{ V}$

over recommended operating free-air temperature range, $V_{CCA} = 2.5\text{ V} \pm 0.2\text{ V}$ (see [Figure 6-1](#))

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CCB}	$T_A = -40^\circ\text{C to } +85^\circ\text{C}$			$T_A = -40^\circ\text{C to } +125^\circ\text{C}$			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
t_{PLH}, t_{PHL}	A	B	$V_{CCB} = 1.2\text{ V}$	2.4			2.4			ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		4.7	0.5		13.5	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		3.9	0.5		12.1	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		3.1	0.5		10.7	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		2.8	0.5		10.2	
t_{PLH}, t_{PHL}	B	A	$V_{CCB} = 1.2\text{ V}$	3			3			ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		4.9	0.5		13.9	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		4	0.5		12.3	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		3.1	0.5		10.7	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		2.9	0.5		10.4	
t_{PZH}, t_{PZL}	$\overline{OE}$	A	$V_{CCB} = 1.2\text{ V}$	2.2			2.2			ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		4.8	0.5		13.7	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		4.8	0.5		13.7	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		4.8	0.5		13.7	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		4.8	0.5		13.7	

5.9 Switching Characteristics, $V_{CCA} = 2.5\text{ V} \pm 0.2\text{ V}$ (continued)

over recommended operating free-air temperature range, $V_{CCA} = 2.5\text{ V} \pm 0.2\text{ V}$ (see [Figure 6-1](#))

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CCB}	$T_A = -40^\circ\text{C to } +85^\circ\text{C}$			$T_A = -40^\circ\text{C to } +125^\circ\text{C}$			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
t_{PZH}, t_{PZL}	$\overline{OE}$	B	$V_{CCB} = 1.2\text{ V}$		4.5			4.5		ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	1.1		7.9	0.5		19.1	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		6.4	0.5		16.5	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		4.6	0.5		13.3	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		4	0.5		12.3	
t_{PHZ}, t_{PLZ}	$\overline{OE}$	A	$V_{CCB} = 1.2\text{ V}$		1.8			1.8		ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		5.1	0.5		14.2	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		5.1	0.5		14.2	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		5.1	0.5		14.2	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		5.1	0.5		14.2	
t_{PHZ}, t_{PLZ}	$\overline{OE}$	B	$V_{CCB} = 1.2\text{ V}$		3.6			3.6		ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		7.1	0.5		17.7	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		6.3	0.5		16.3	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		5.1	0.5		14.2	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		3.9	0.5		12.1	

5.10 Switching Characteristics, $V_{CCA} = 3.3\text{ V} \pm 0.3\text{ V}$

over recommended operating free-air temperature range, $V_{CCA} = 3.3\text{ V} \pm 0.3\text{ V}$ (see [Figure 6-1](#))

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CCB}	$T_A = -40^\circ\text{C to } +85^\circ\text{C}$			$T_A = -40^\circ\text{C to } +125^\circ\text{C}$			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
t_{PLH}, t_{PHL}	A	B	$V_{CCB} = 1.2\text{ V}$		2.3			2.3		ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		4.5	0.5		13.2	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	t_{PLH}	0.5	3.7	0.5		11.1	
				t_{PHL}	0.5	3.3	0.5		11.1	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		2.9	0.5		10.4	
t_{PLH}, t_{PHL}	B	A	$V_{CCB} = 1.2\text{ V}$		3.5			3.5		ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		6.8	0.5		17.2	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		3.9	0.5		12.1	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		2.8	0.5		10.2	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		2.5	0.5		9.7	
t_{PZH}, t_{PZL}	$\overline{OE}$	A	$V_{CCB} = 1.2\text{ V}$		2			2		ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		4	0.5		12.3	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		4	0.5		12.3	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		4	0.5		12.3	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		4	0.5		12.3	
t_{PZH}, t_{PZL}	$\overline{OE}$	B	$V_{CCB} = 1.2\text{ V}$		4.5			4.5		ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	1.1		7.8	0.5		18.9	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		6.2	0.5		16.1	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		4.5	0.5		13.2	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		3.9	0.5		12.3	

5.10 Switching Characteristics, $V_{CCA} = 3.3\text{ V} \pm 0.3\text{ V}$ (continued)

over recommended operating free-air temperature range, $V_{CCA} = 3.3\text{ V} \pm 0.3\text{ V}$ (see Figure 6-1)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CCB}	$T_A = -40^\circ\text{C to } +85^\circ\text{C}$			$T_A = -40^\circ\text{C to } +125^\circ\text{C}$			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
t_{PHZ}, t_{PLZ}	$\overline{OE}$	A	$V_{CCB} = 1.2\text{ V}$		1.7			1.7		ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		4	0.5		12.3	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		4	0.5		12.3	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		4	0.5		12.3	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		4	0.5		12.3	
t_{PHZ}, t_{PLZ}	$\overline{OE}$	B	$V_{CCB} = 1.2\text{ V}$		3.4			3.4		ns
			$V_{CCB} = 1.5\text{ V} \pm 0.1\text{ V}$	0.5		6.9	0.5		17.4	
			$V_{CCB} = 1.8\text{ V} \pm 0.15\text{ V}$	0.5		6	0.5		15.8	
			$V_{CCB} = 2.5\text{ V} \pm 0.2\text{ V}$	0.5		4.8	0.5		13.7	
			$V_{CCB} = 3.3\text{ V} \pm 0.3\text{ V}$	0.5		4.2	0.5		12.6	

5.11 Operating Characteristics

$T_A = 25^\circ\text{C}$

PARAMETER			TEST CONDITIONS	$V_{CCA} =$ $V_{CCB} = 1.2\text{ V}$	$V_{CCA} =$ $V_{CCB} = 1.5\text{ V}$	$V_{CCA} =$ $V_{CCB} = 1.8\text{ V}$	$V_{CCA} =$ $V_{CCB} = 2.5\text{ V}$	$V_{CCA} =$ $V_{CCB} = 3.3\text{ V}$	UNIT
				TYP	TYP	TYP	TYP	TYP	
C_{pdA} (1)	A to B	Outputs enabled	$C_L = 0,$ $f = 10\text{ MHz},$ $t_r = t_f = 1\text{ ns}$	1	1	1	1	1	pF
		Outputs disabled		1	1	1	1	1	
	B to A	Outputs enabled		12	12	12	13	14	
		Outputs disabled		1	1	1	1	1	
C_{pdB} (1)	A to B	Outputs enabled	$C_L = 0,$ $f = 10\text{ MHz},$ $t_r = t_f = 1\text{ ns}$	12	12	12	13	14	pF
		Outputs disabled		1	1	1	1	1	
	B to A	Outputs enabled		1	1	1	1	1	
		Outputs disabled		1	1	1	1	1	

(1) Power dissipation capacitance per transceiver

5.12 Typical Total Static Power Consumption ($I_{CCA} + I_{CCB}$)

V_{CCB}	V_{CCA}						UNIT
	0 V	1.2 V	1.5 V	1.8 V	2.5 V	3.3 V	
0 V	0	<0.5	<0.5	<0.5	<0.5	<0.5	μA
1.2 V	<0.5	<1	<1	<1	<1	1	
1.5 V	<0.5	<1	<1	<1	<1	1	
1.8 V	<0.5	<1	<1	<1	<1	<1	
2.5 V	<0.5	1	<1	<1	<1	<1	
3.3 V	<0.5	1	<1	<1	<1	<1	

5.13 Typical Characteristics

$T_A = 25^\circ\text{C}$

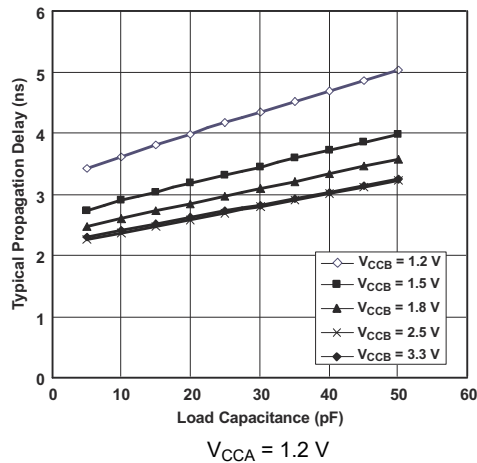


Figure 5-1. Typical Propagation Delay (A to B) vs Load Capacitance

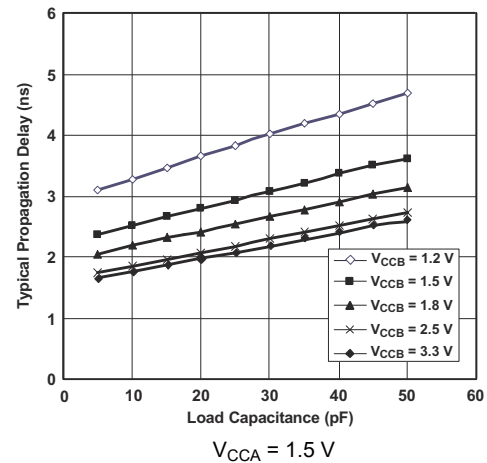


Figure 5-2. Typical Propagation Delay (A to B) vs Load Capacitance

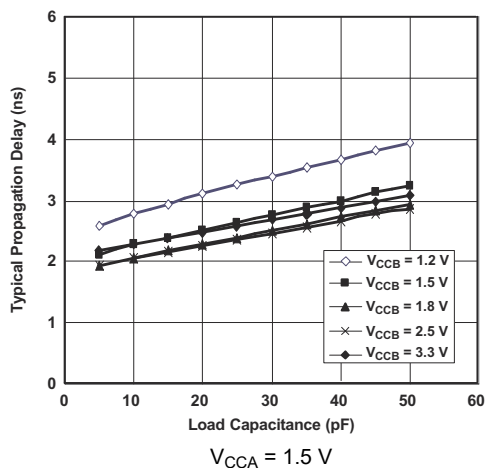


Figure 5-3. Typical Propagation Delay (A to B) vs Load Capacitance

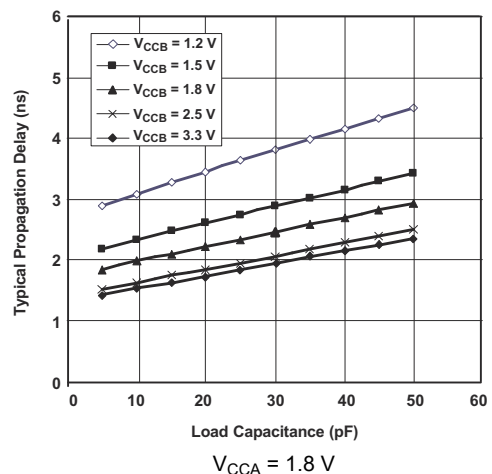


Figure 5-4. Typical Propagation Delay (A to B) vs Load Capacitance

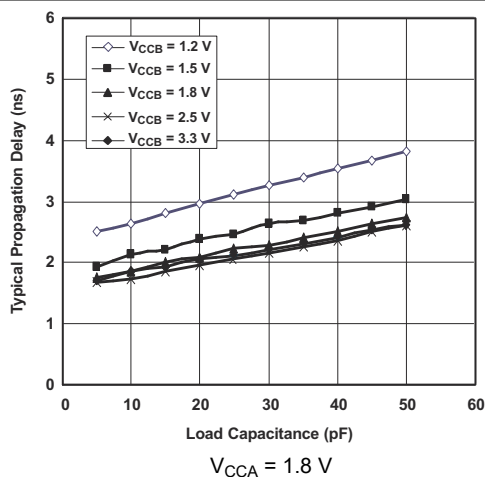


Figure 5-5. Typical Propagation Delay (A to B) vs Load Capacitance

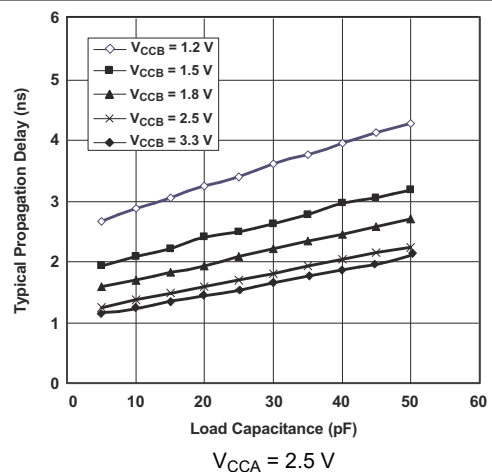


Figure 5-6. Typical Propagation Delay (A to B) vs Load Capacitance

5.13 Typical Characteristics (continued)

$T_A = 25^\circ\text{C}$

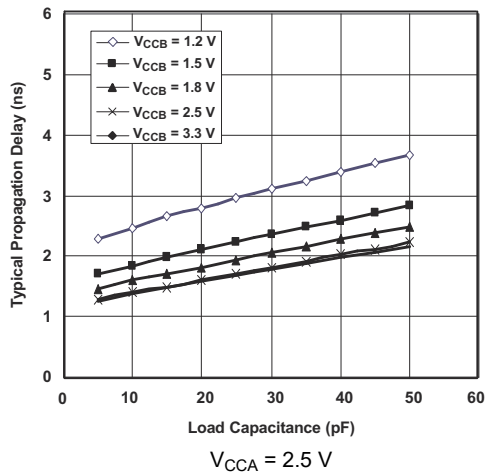


Figure 5-7. Typical Propagation Delay (A to B) vs Load Capacitance

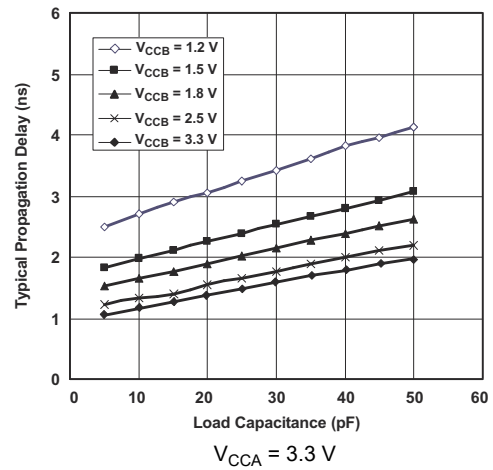


Figure 5-8. Typical Propagation Delay (A to B) vs Load Capacitance

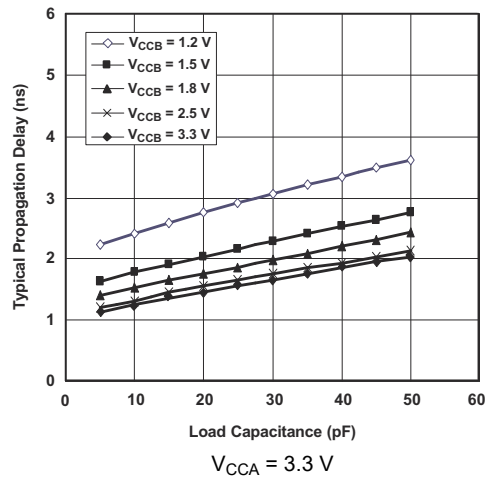
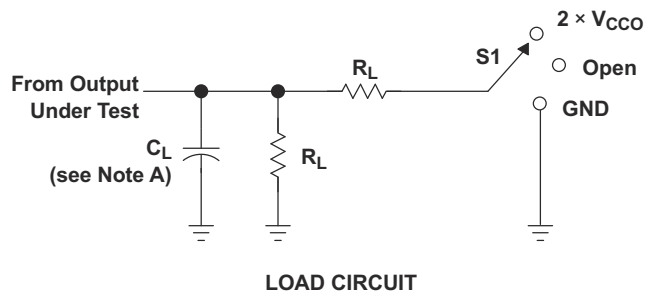


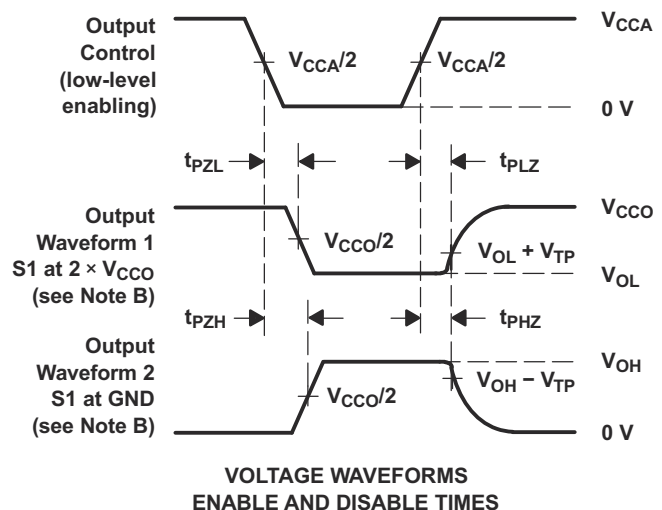
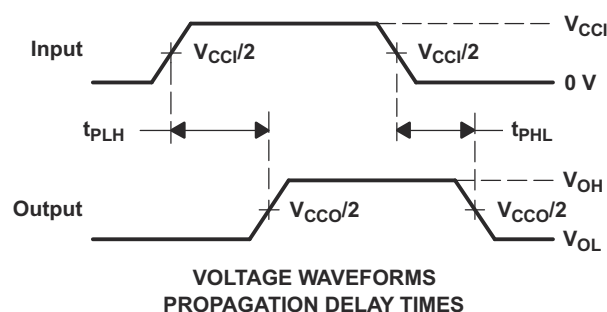
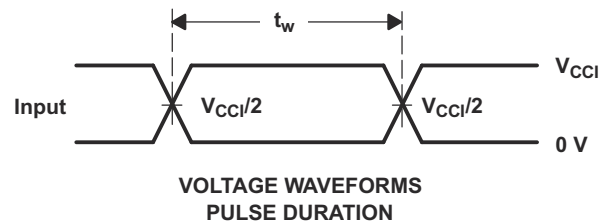
Figure 5-9. Typical Propagation Delay (A to B) vs Load Capacitance

Parameter Measurement Information



V_{CCO}	C_L	R_L	V_{TP}
1.2 V	15 pF	2 kW	0.1 V
1.5 V $\pm$ 0.1 V	15 pF	2 kW	0.1 V
1.8 V $\pm$ 0.15 V	15 pF	2 kW	0.15 V
2.5 V $\pm$ 0.2 V	15 pF	2 kW	0.15 V
3.3 V $\pm$ 0.3 V	15 pF	2 kW	0.3 V

TEST	S1
t_{pd}	Open
t_{PLZ}/t_{PZL}	$2 \times V_{CCO}$
t_{PHZ}/t_{PZH}	GND



- NOTES:
- A. C_L includes probe and jig capacitance.
 - B. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high, except when disabled by the output control.
 - C. All input pulses are supplied by generators having the following characteristics: PRR = 10 MHz, $Z_O = 50 \Omega$, $dv/dt \geq 1$ V/ns.
 - D. The outputs are measured one at a time, with one transition per measurement.
 - E. t_{PLZ} and t_{PHZ} are the same as t_{dis} .
 - F. t_{PZL} and t_{PZH} are the same as t_{en} .
 - G. t_{PLH} and t_{PHL} are the same as t_{pd} .
 - H. V_{CCI} is the V_{CC} associated with the input port.
 - I. V_{CCO} is the V_{CC} associated with the output port.

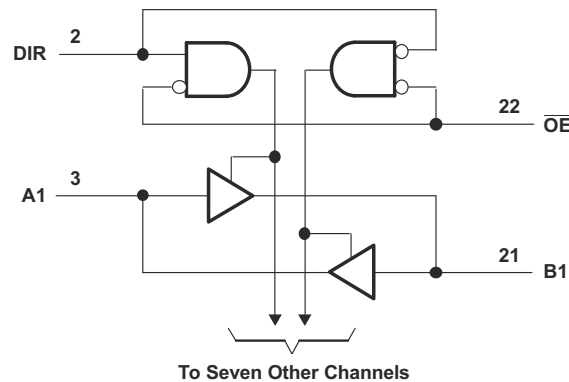
Figure 6-1. Load Circuit and Voltage Waveforms

6 Detailed Description

6.1 Overview

The SN74AVC8T245 is an 8-bit, dual-supply noninverting transceiver with bidirectional voltage level translation. V_{CCA} supports pins A and the control pins (DIR and $\overline{OE}$), and V_{CCB} supports pins B. The A port is able to accept I/O voltages ranging from 1.2 V to 3.6 V, while the B port can accept I/O voltages from 1.2 V to 3.6 V. A high on DIR allows data transmission from A to B and a low on DIR allows data transmission from B to A when $\overline{OE}$ is set to low. When $\overline{OE}$ is set to high, both A and B are in the high-impedance state.

6.2 Functional Block Diagram



6.3 Feature Description

6.3.1 Fully Configurable Dual-Rail Design

The fully configurable dual-rail design allows each port to operate over the full 1.2-V to 3.6-V power-supply range. Both V_{CCA} and V_{CCB} can be supplied at any voltage between 1.2 V and 3.6 V making the device an excellent choice for translating between any of the low voltage nodes (1.2 V, 1.8 V, 2.5 V, and 3.3 V).

6.3.2 Support High-Speed Translation

SN74AVC8T245 can support high data rate application. The translated signal data rate can be up to 320Mbps when the device power supply is more than 1.8 V.

6.3.3 I_{off} Supports Partial-Power-Down Mode Operation

I_{off} prevents backflow current by disabling I/O output circuits when device is in partial power-down mode. The inputs and outputs for this device enter a high-impedance state when the device is powered down, inhibiting current backflow into the device. The maximum leakage into or out of any input or output pin on the device is specified by I_{off} in the *Electrical Characteristics*.

6.3.4 Balanced High-Drive CMOS Push-Pull Outputs

A balanced output allows the device to sink and source similar currents. The high drive capability of this device creates fast edges into light loads, so routing and load conditions should be considered to prevent ringing. Additionally, the outputs of this device are capable of driving larger currents than the device can sustain without being damaged. Two outputs can be connected together for 2X stronger output drive strength. The electrical and thermal limits defined in the *Absolute Maximum Ratings* must be followed at all times.

6.3.5 Vcc Isolation

The I/Os of both ports will enter a high-impedance state when one of the supplies are at GND, while the other supply is still connected to the device (IOZ shown in *Electrical Characteristics*).

6.4 Device Functional Modes

The SN74AVC8T245 is a voltage level transceiver that can operate from 1.2 V to 3.6 V (V_{CCA}) and 1.2 V to 3.6 V (V_{CCB}). The signal translation between 1.2 V and 3.6 V requires direction control and output enable control. When $\overline{OE}$ is low and DIR is high, data transmission is from A to B. When $\overline{OE}$ is low and DIR is low, data transmission is from B to A. When $\overline{OE}$ is high, both output ports will be high-impedance.

**Table 6-1. Function Table
(Each 8-Bit Section)**

INPUTS		OPERATION
$\overline{OE}$	DIR	
L	L	B data to A bus
L	H	A data to B bus
H	X	All outputs Hi-Z

7 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

7.1 Application Information

The SN74AVC8T245 device can be used in level-translation applications for interfacing devices or systems operating at different interface voltages with one another. The SN74AVC8T245 device is an excellent choice for data transmission when direction is different. It is recommended to tie all unused I/Os to GND. The device should not have any floating I/Os when changing translation direction. The maximum data rate can be up to 320Mbps when device voltage power supply is more than 1.8 V.

7.2 Typical Application

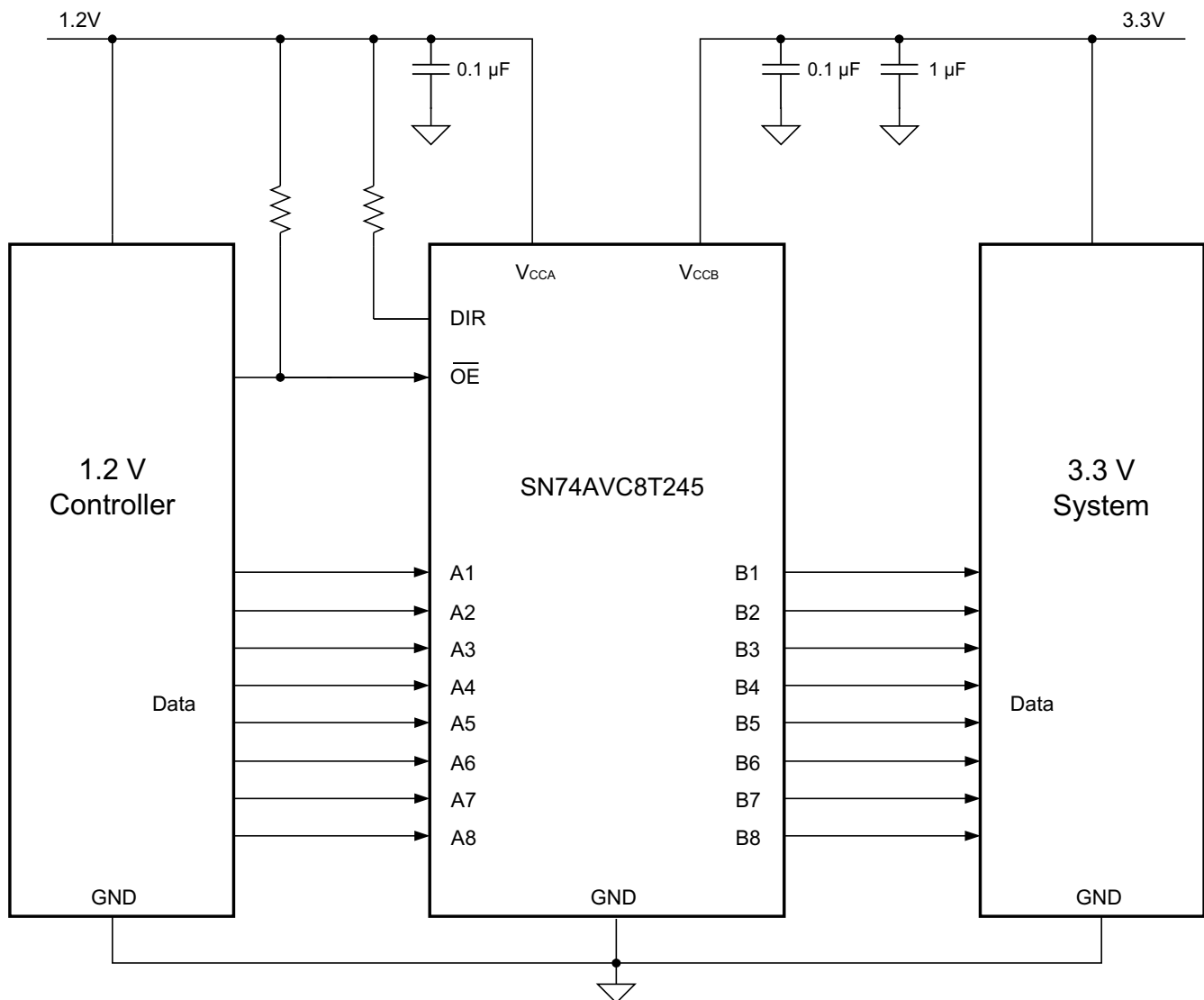


Figure 7-1. Typical Application Schematic

7.2.1 Design Requirements

For this design example, use the parameters listed in [Table 7-1](#).

Table 7-1. Design Parameters

DESIGN PARAMETERS	EXAMPLE VALUE
Input voltage range	1.2 V to 3.6 V
Output voltage range	1.2 V to 3.6 V

7.2.2 Detailed Design Procedure

To begin the design process, determine the following:

- Input voltage range
 - Use the supply voltage of the device that is driving the SN74AVC8T245 device to determine the input voltage range. For a valid logic high the value must exceed the V_{IH} of the input port. For a valid logic low the value must be less than the V_{IL} of the input port.
- Output voltage range
 - Use the supply voltage of the device that the SN74AVC8T245 device is driving to determine the output voltage range.

7.2.3 Application Curve

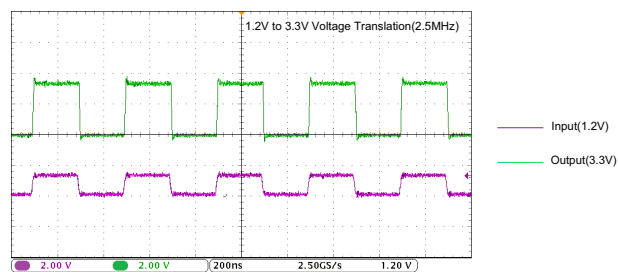


Figure 7-2. Translation Up (1.2 V to 3.3 V) at 2.5 MHz

7.3 Power Supply Recommendations

The SN74AVC8T245 device uses two separate configurable power-supply rails, V_{CCA} and V_{CCB} . V_{CCA} accepts any supply voltage from 1.2 V to 3.6 V and V_{CCB} accepts any supply voltage from 1.2 V to 3.6 V. The A port and B port are designed to track V_{CCA} and V_{CCB} , respectively, allowing for low-voltage bidirectional translation between any of the 1.2-V, 1.5-V, 1.8-V, 2.5-V and 3.3-V voltage nodes. The recommendation is to first power-up the input supply rail to help avoid internal floating while the output supply rail ramps up. However, both power-supply rails can be ramped up simultaneously.

The output-enable $\overline{OE}$ input circuit is designed so that it is supplied by V_{CCA} and when the $\overline{OE}$ input is high, all outputs are placed in the high-impedance state. To ensure the high-impedance state of the outputs during power up or power down, the $\overline{OE}$ input pin must be tied to V_{CCA} through a pullup resistor and must not be enabled until V_{CCA} and V_{CCB} are fully ramped and stable. The minimum value of the pullup resistor to V_{CCA} is determined by the current-sinking capability of the driver.

7.4 Layout

7.4.1 Layout Guidelines

For device reliability, follow common printed-circuit board layout guidelines such as:

- Use bypass capacitors on power supplies.
- Use short trace lengths to avoid excessive loading.
- Place pads on the signal paths for loading capacitors or pullup resistors to adjust the signals rise and fall times, depending on the system requirements.

7.4.2 Layout Example

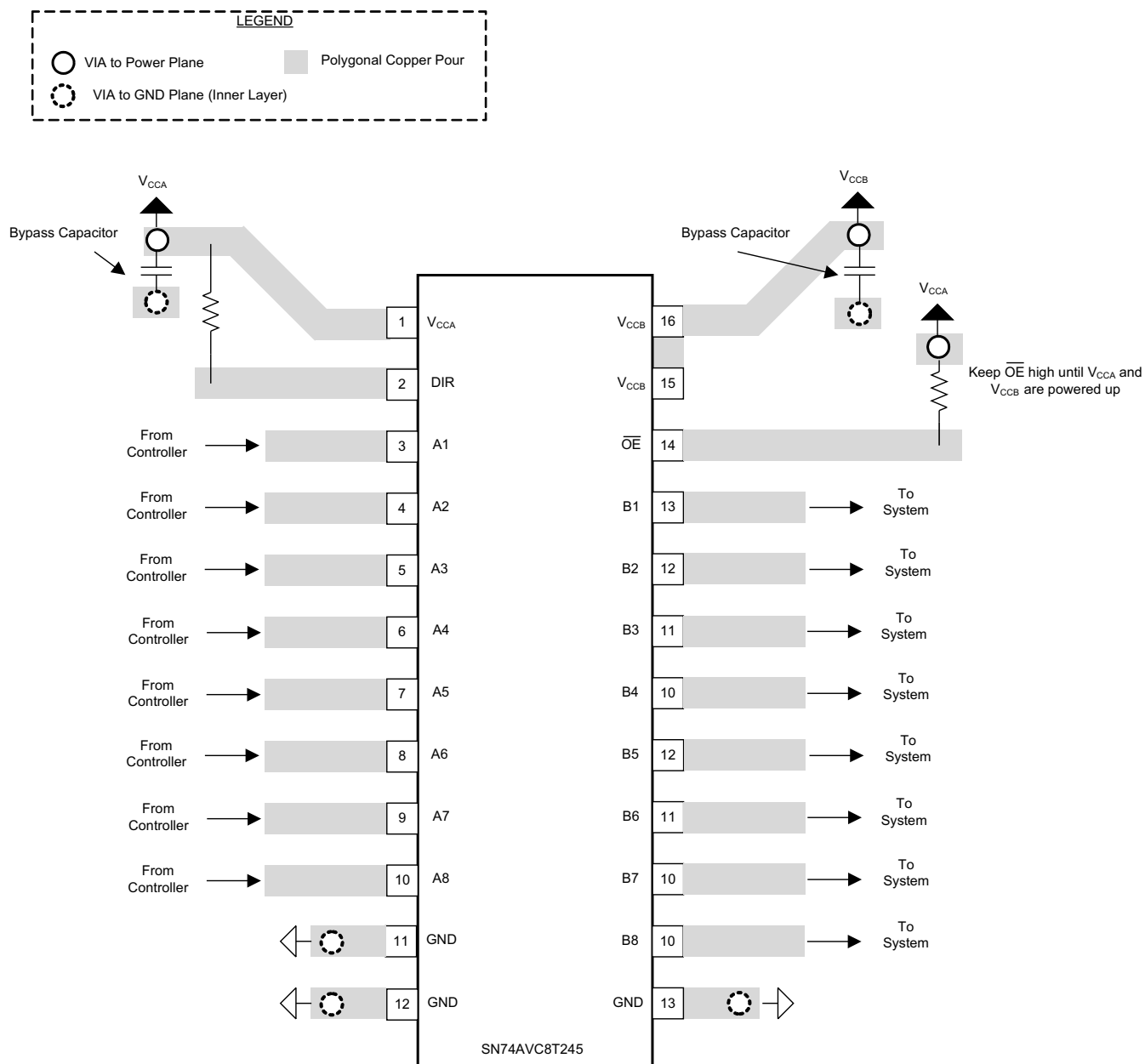


Figure 7-3. SN74AVC8T245 Layout Example

8 Device and Documentation Support

8.1 Documentation Support

8.1.1 Related Documentation

For related documentation, see the following:

- Texas Instruments, [Implications of Slow or Floating CMOS Inputs application note](#)

8.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

8.3 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

8.4 Trademarks

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8.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

8.6 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

9 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision J (March 2017) to Revision K (November 2023)	Page
• Updated the numbering format for tables, figures, and cross-references throughout the document.....	1
• Updated the <i>Package Information</i> table to include package lead size.....	1
• Updated the <i>Thermal Information</i> table for all packages.....	6
Changes from Revision I (December 2014) to Revision J (March 2017)	Page
• Changed MAX value for Operating free-air temperature, T_A from: 85°C to: 125°C.....	5
• Added values for $T_A = -40^\circ\text{C}$ to $+125^\circ\text{C}$ in <i>Electrical Characteristics</i> and all <i>Switching Characteristics</i> tables.....	6
• Added <i>Documentation Support</i> section, <i>Receiving Notification of Documentation Updates</i> , and <i>Community Resources</i> section.....	20
Changes from Revision H (February 2007) to Revision I (December 2014)	Page
• Added <i>Pin Configuration and Functions</i> section, <i>ESD Rating</i> table, <i>Feature Description</i> section, <i>Device Functional Modes</i> , <i>Application and Implementation</i> section, <i>Power Supply Recommendations</i> section, <i>Layout</i> section, <i>Device and Documentation Support</i> section, and <i>Mechanical, Packaging, and Orderable Information</i> section	1

10 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
74AVC8T245RHRLRG4	Active	Production	VQFN (RHL) 24	1000 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	WE245
SN74AVC8T245DGVR	Active	Production	TVSOP (DGV) 24	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245DGVR.A	Active	Production	TVSOP (DGV) 24	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245DGVR.B	Active	Production	TVSOP (DGV) 24	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245DGVRG4	Active	Production	TVSOP (DGV) 24	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245DGVRG4.A	Active	Production	TVSOP (DGV) 24	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245DGVRG4.B	Active	Production	TVSOP (DGV) 24	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245PW	Active	Production	TSSOP (PW) 24	60 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245PW.B	Active	Production	TSSOP (PW) 24	60 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245PWE4	Active	Production	TSSOP (PW) 24	60 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245PWG4	Active	Production	TSSOP (PW) 24	60 TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245PWR	Active	Production	TSSOP (PW) 24	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245PWR.A	Active	Production	TSSOP (PW) 24	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245PWR.B	Active	Production	TSSOP (PW) 24	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245PWRE4	Active	Production	TSSOP (PW) 24	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245PWRG4	Active	Production	TSSOP (PW) 24	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	WE245
SN74AVC8T245RHRLR	Active	Production	VQFN (RHL) 24	1000 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	WE245
SN74AVC8T245RHRLR.A	Active	Production	VQFN (RHL) 24	1000 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	WE245
SN74AVC8T245RHRLR.B	Active	Production	VQFN (RHL) 24	1000 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	WE245

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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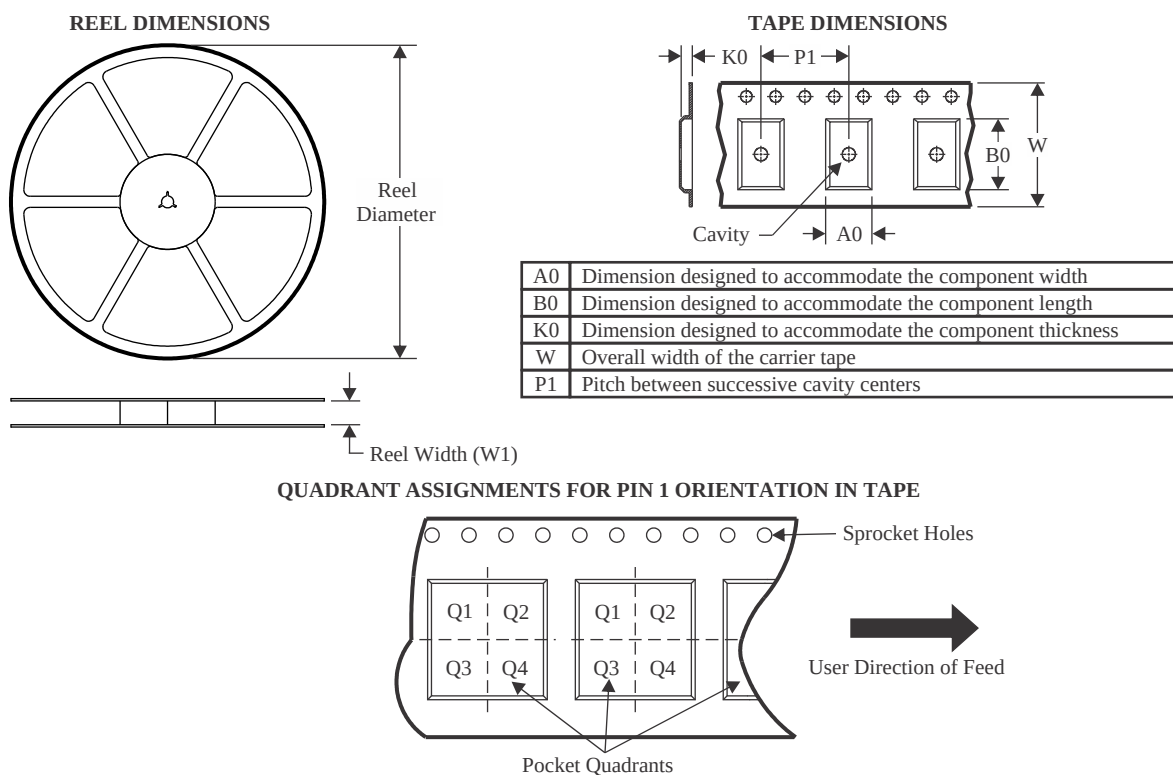
OTHER QUALIFIED VERSIONS OF SN74AVC8T245 :

- Automotive : [SN74AVC8T245-Q1](#)

NOTE: Qualified Version Definitions:

- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects

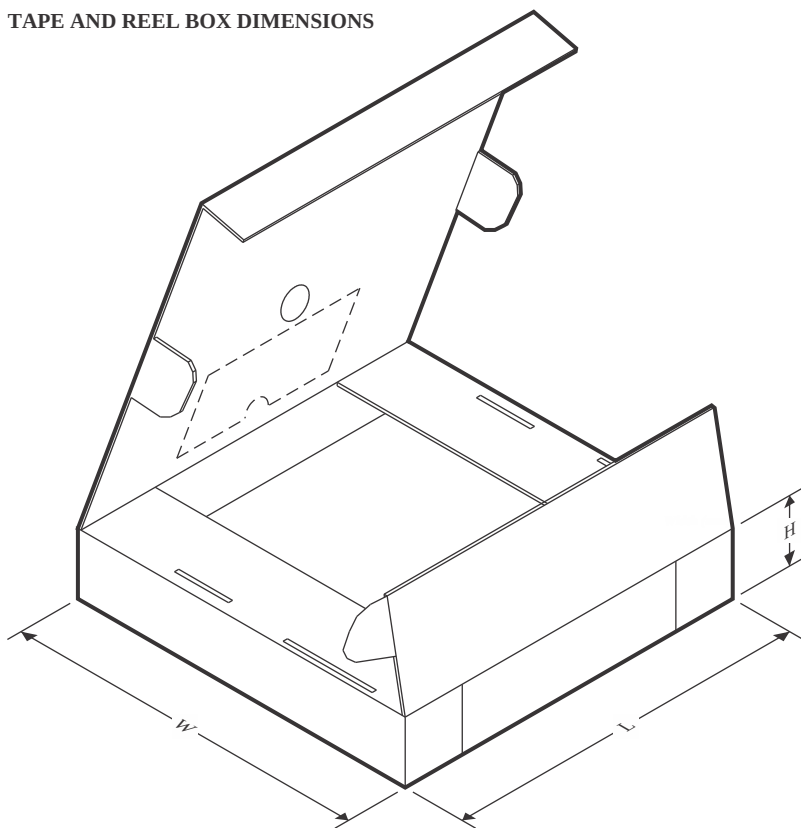
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN74AVC8T245DGVR	TVSOP	DGV	24	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
SN74AVC8T245DGVRG4	TVSOP	DGV	24	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
SN74AVC8T245PWR	TSSOP	PW	24	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
SN74AVC8T245RHLLR	VQFN	RHL	24	1000	330.0	12.4	3.8	5.8	1.2	8.0	12.0	Q1
SN74AVC8T245RHLLR	VQFN	RHL	24	1000	180.0	12.4	3.8	5.8	1.2	8.0	12.0	Q1
SN74AVC8T245RHLLR	VQFN	RHL	24	1000	180.0	12.4	3.8	5.8	1.2	8.0	12.0	Q1

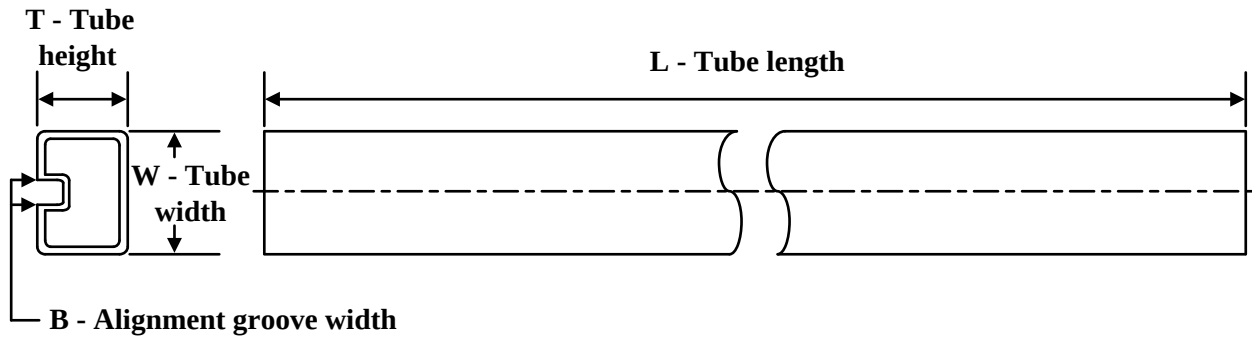
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

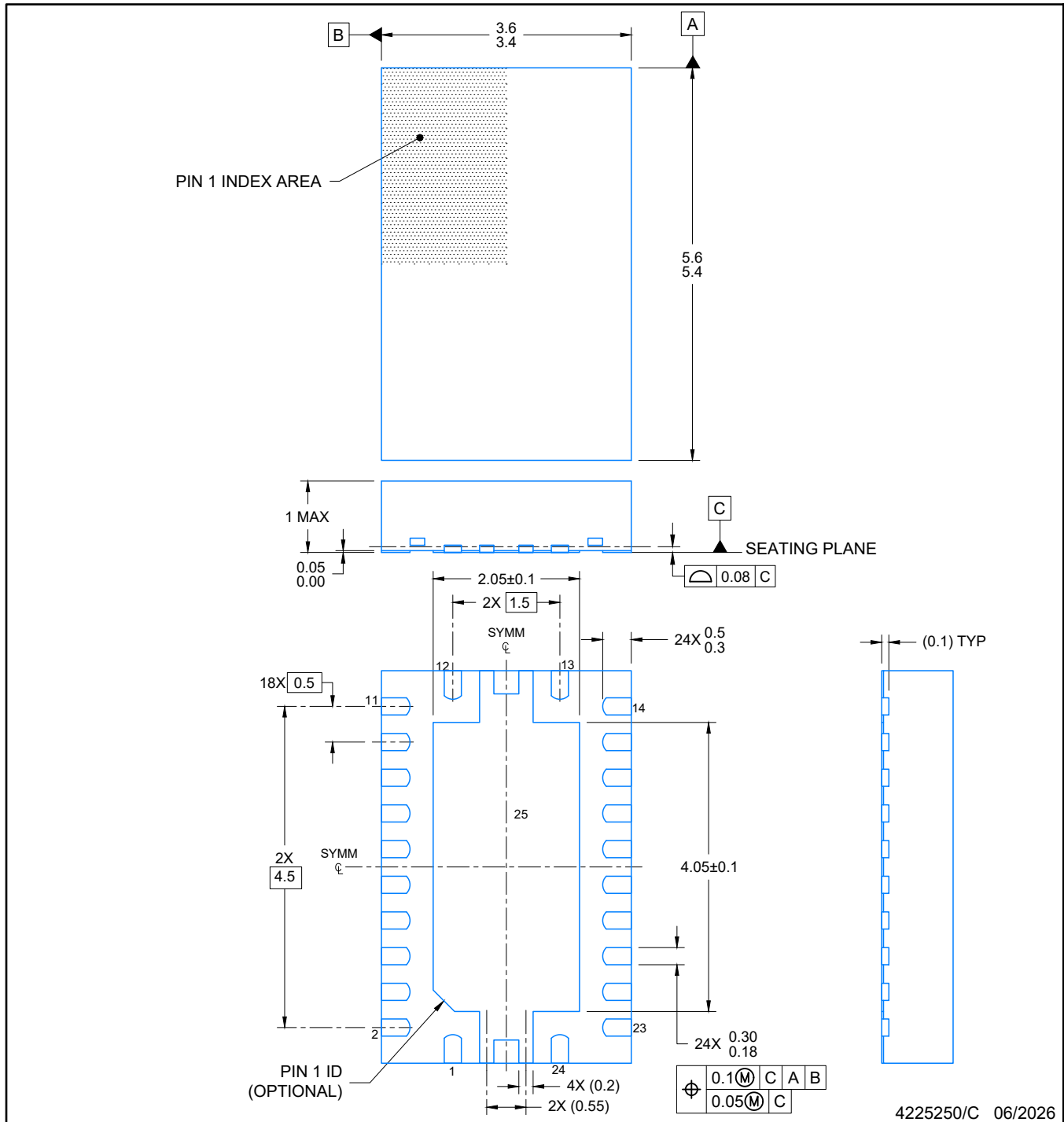
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN74AVC8T245DGVR	TVSOP	DGV	24	2000	353.0	353.0	32.0
SN74AVC8T245DGVRG4	TVSOP	DGV	24	2000	353.0	353.0	32.0
SN74AVC8T245PWR	TSSOP	PW	24	2000	353.0	353.0	32.0
SN74AVC8T245RHLR	VQFN	RHL	24	1000	367.0	367.0	35.0
SN74AVC8T245RHLR	VQFN	RHL	24	1000	213.0	191.0	35.0
SN74AVC8T245RHLR	VQFN	RHL	24	1000	210.0	185.0	35.0

TUBE

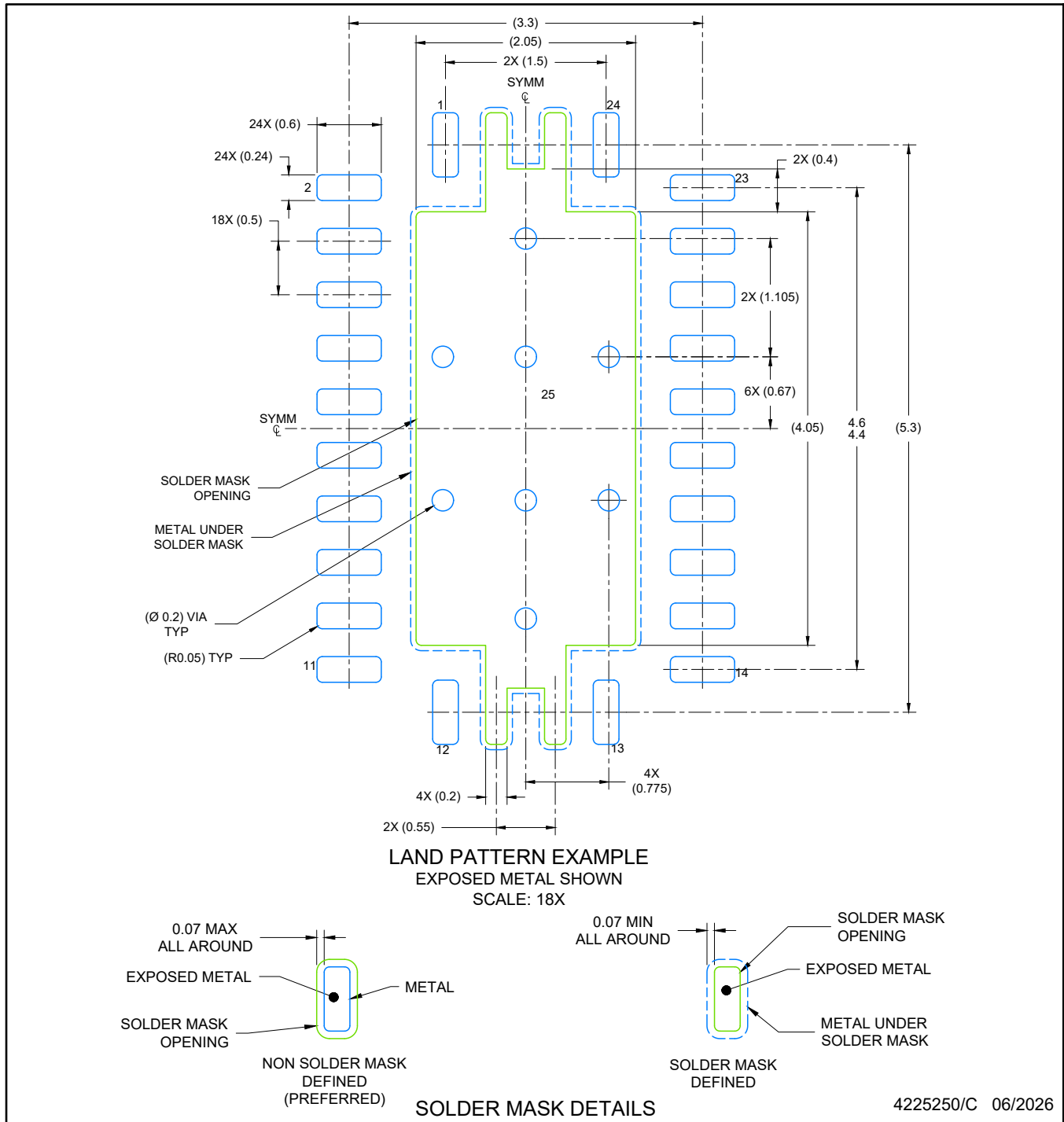


*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
SN74AVC8T245PW	PW	TSSOP	24	60	530	10.2	3600	3.5
SN74AVC8T245PW.B	PW	TSSOP	24	60	530	10.2	3600	3.5
SN74AVC8T245PWE4	PW	TSSOP	24	60	530	10.2	3600	3.5
SN74AVC8T245PWG4	PW	TSSOP	24	60	530	10.2	3600	3.5

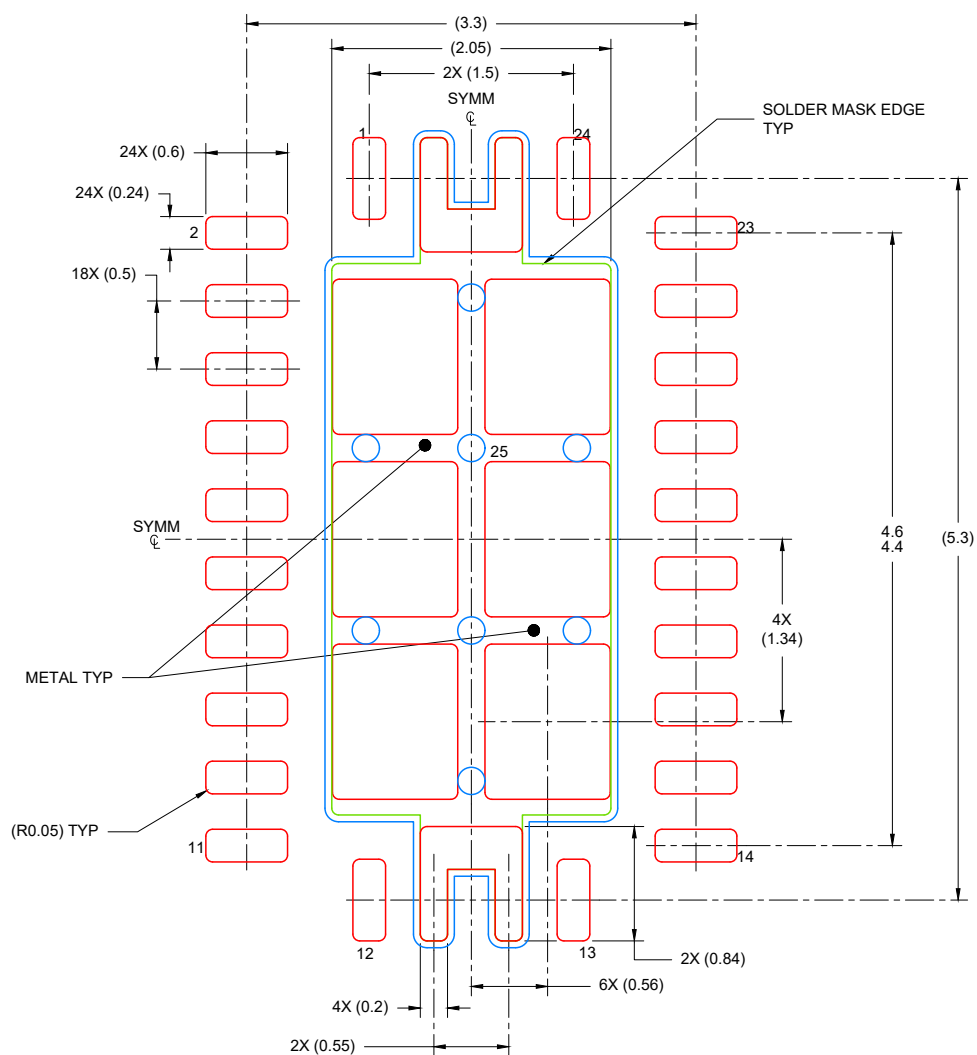


4225250/C 06/2026



NOTES: (continued)

- This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature number SLUA271 (www.ti.com/lit/sluea271).
- Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL

EXPOSED PAD
80% PRINTED COVERAGE BY AREA
SCALE: 18X

4225250/C 06/2026

NOTES: (continued)

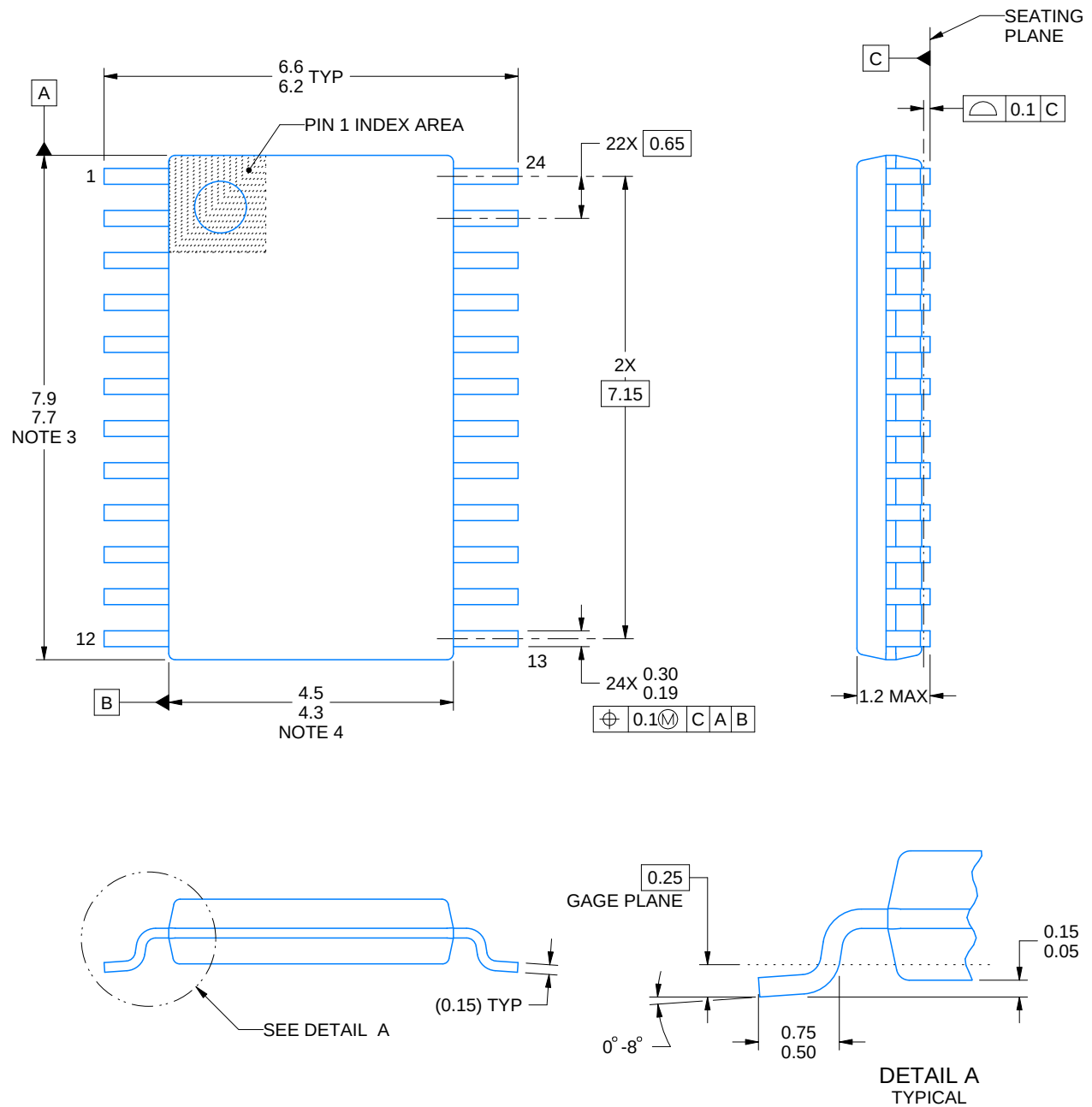
6. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

PW0024A

PACKAGE OUTLINE

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



4220208/A 02/2017

NOTES:

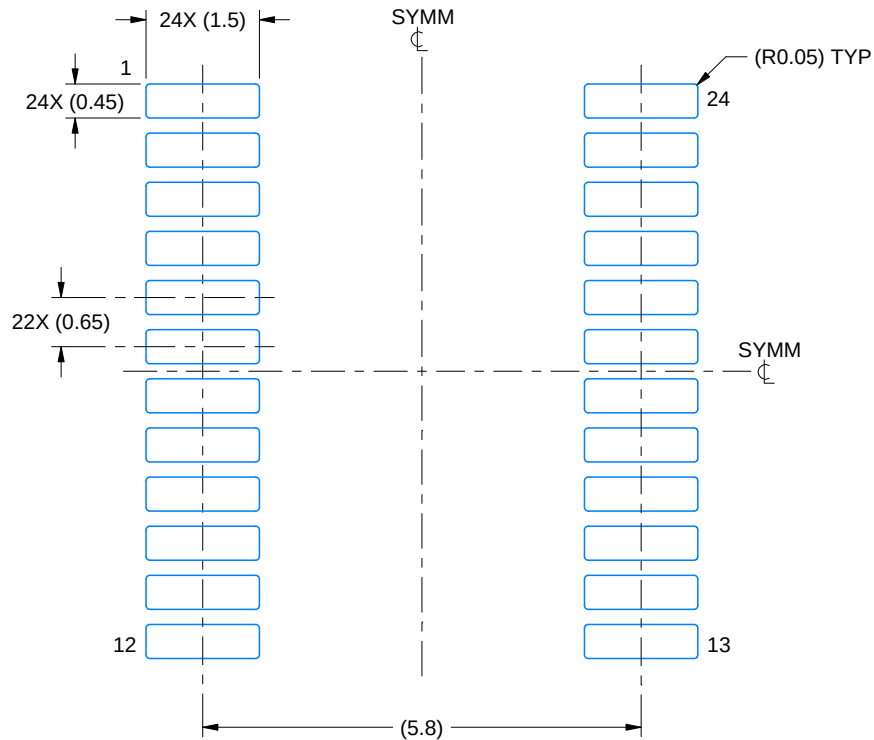
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.

EXAMPLE BOARD LAYOUT

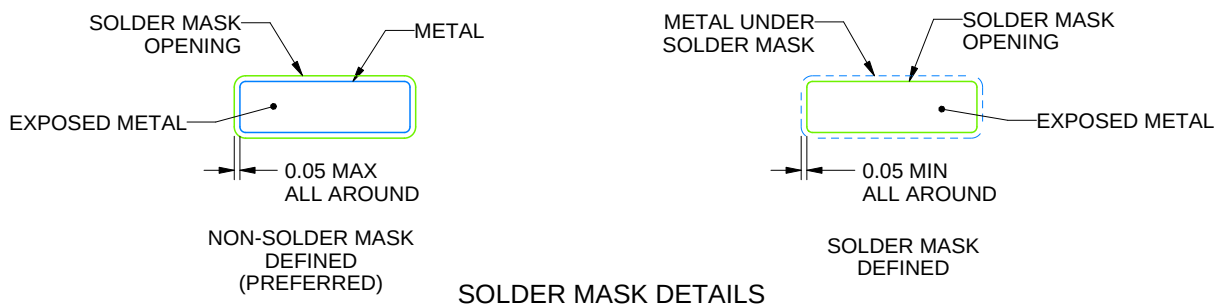
PW0024A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X



4220208/A 02/2017

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

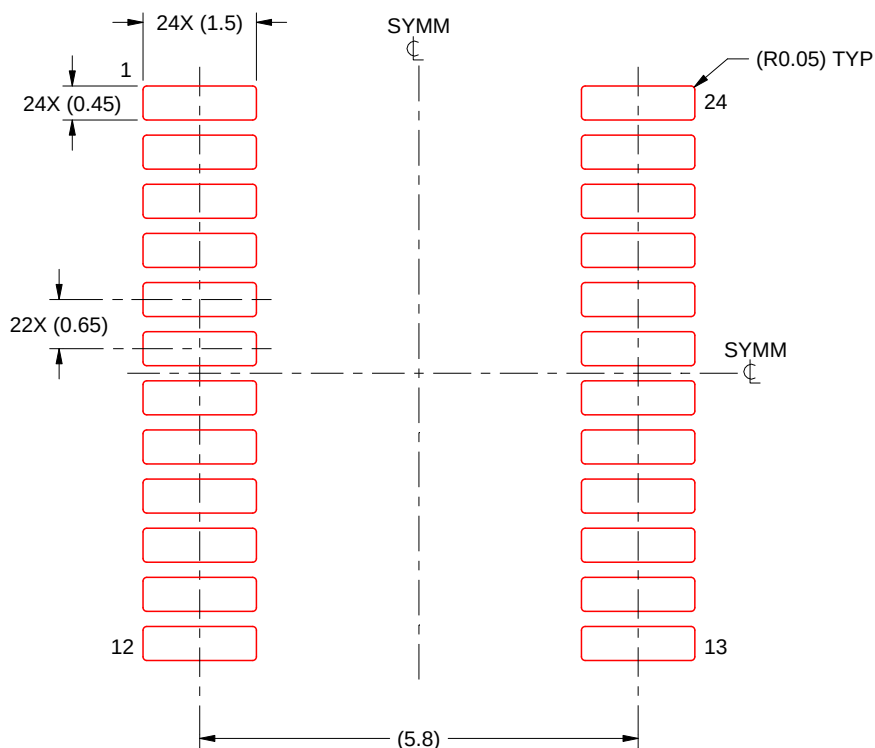
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

PW0024A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE

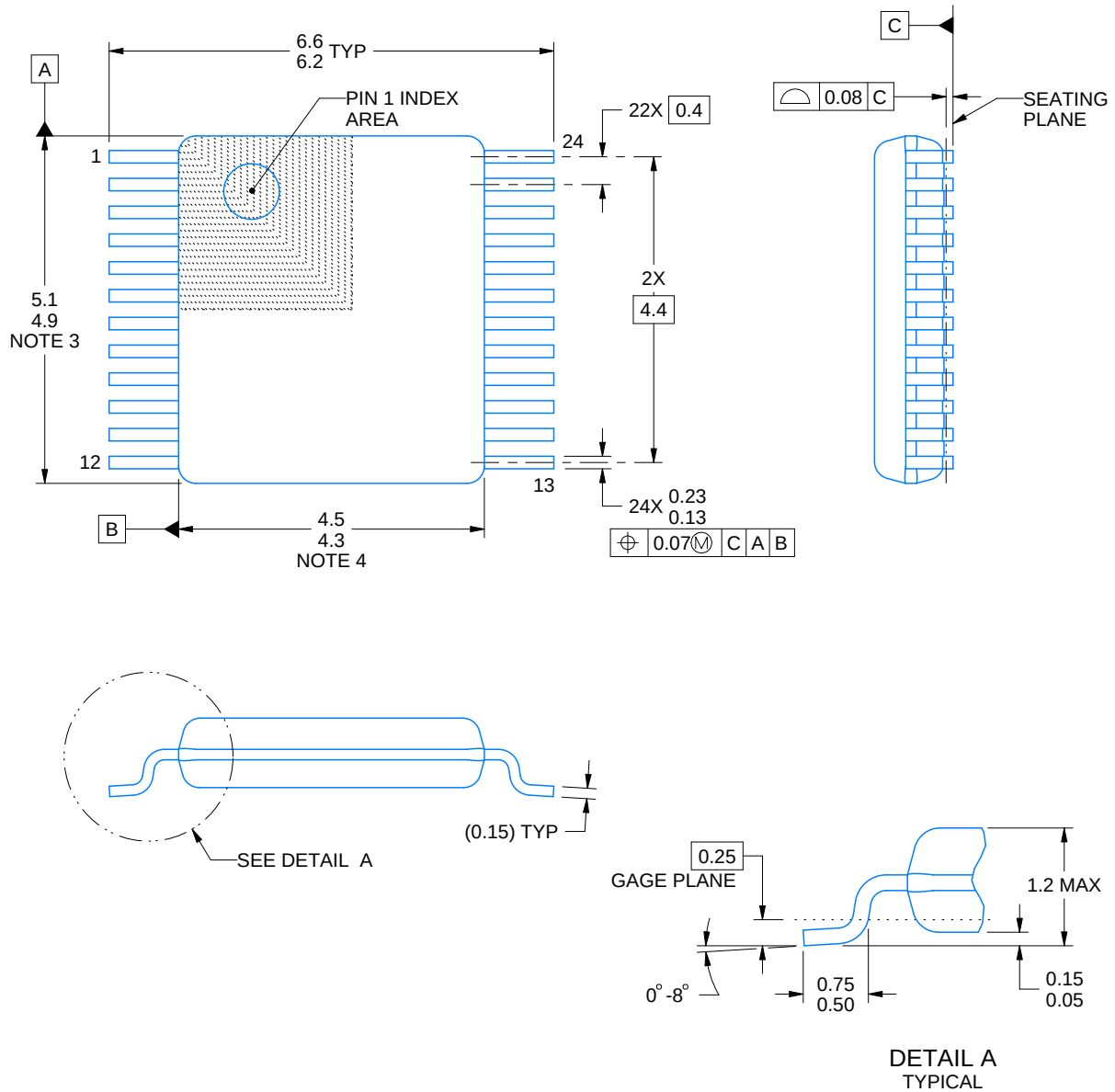
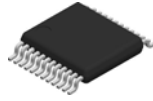


SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

4220208/A 02/2017

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.



4229221/A 12/2022

NOTES:

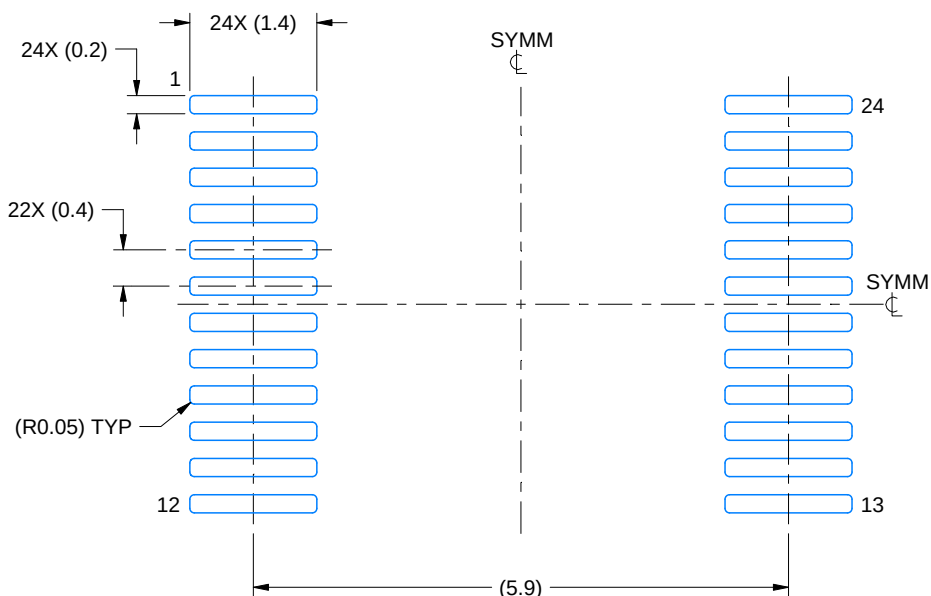
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.

EXAMPLE BOARD LAYOUT

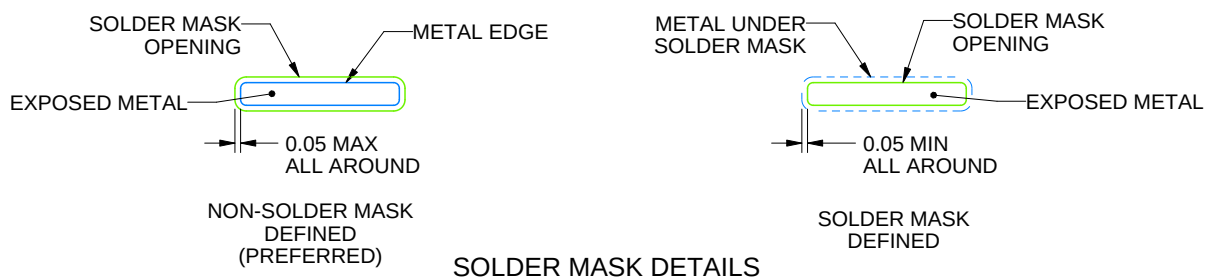
DGV0024A

TVSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 12X



SOLDER MASK DETAILS

4229221/A 12/2022

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

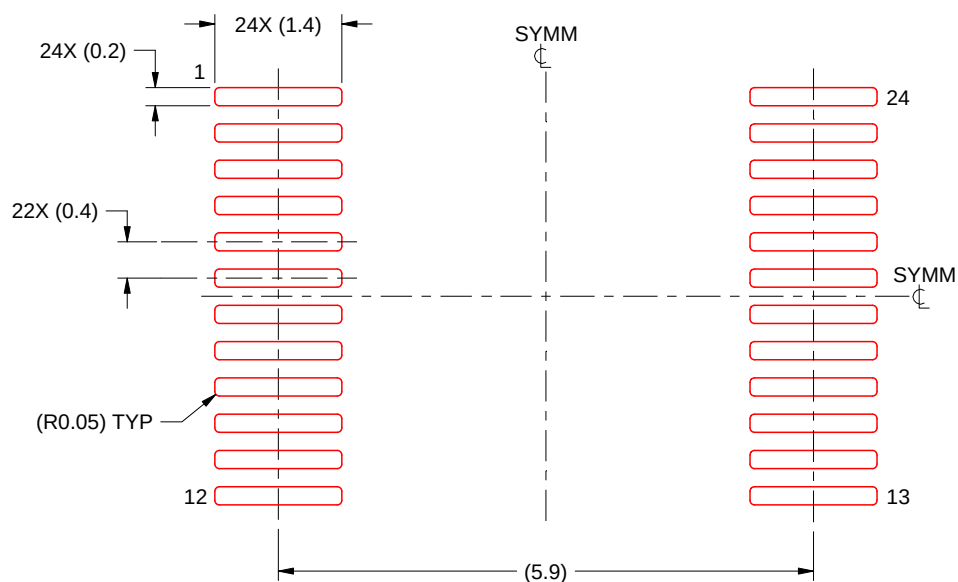
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

DGV0024A

TVSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 12X

4229221/A 12/2022

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

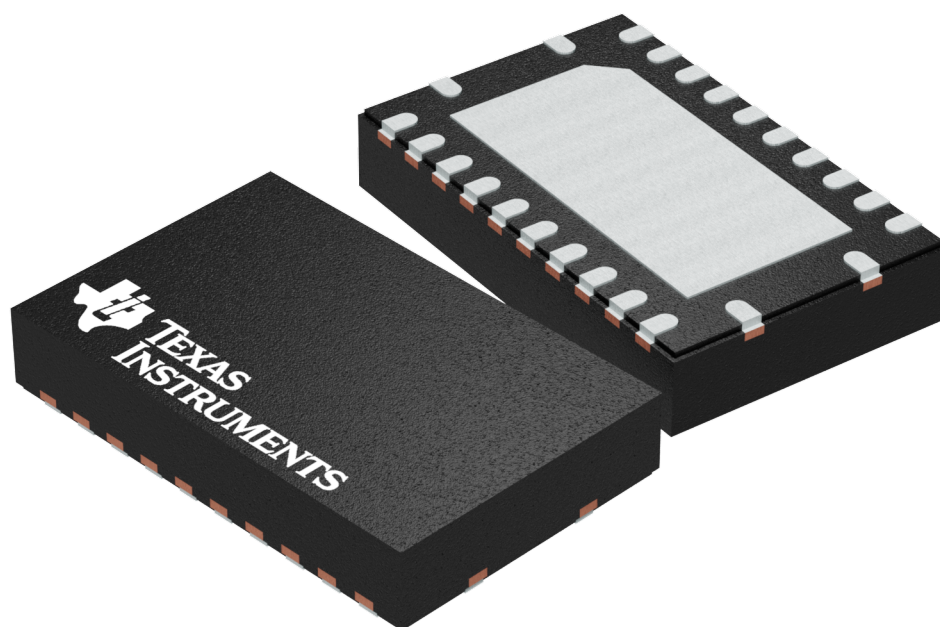
GENERIC PACKAGE VIEW

RGY 24

VQFN - 1 mm max height

5.5 x 3.5 mm, 0.5 mm pitch

PLASTIC QUAD FLATPACK - NO LEAD



Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

4203539-5/J

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